EQN

ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

Kinesthetic Quadratics, Factoring & Polynomial Binomial Arrays

STUDIO CONTINUUM: 1·Ten-Frames · 2·Dominoes · 3·Clock · 4·Fractions · 5·Primes · **6·Algebra** · 7·Probability · 8·Volvelle · 9·Quest · 10·Rack · 11·SC-32

CARD 06 / 10 • ALGB-01

MATHEMATICAL REFERENCE

1. WHAT IS THIS MANIPULATIVE?

GEOMETRIC AREA MODEL. MULTIPLY BY DIMENSION.

Quadratic equations model geometric area.
Polynomial tiles model multiplication geometrically:
Area = Length × Width = (x + a)(x + b)
Arranging tiles into a rectangle factors polynomials!

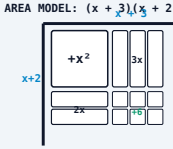
CALIBRATED POLYNOMIAL DATUM

- Unit Datum: +1 Constant Tile = 1U = 7.00 mm pitch
- Linear Rod: +x Tile = 5U × 1U = 35.00 × 7.00 mm
- Quadratic: +x² Tile = 5U × 5U = 35.00 × 35.00 mm
- Connector Spec: 1.85mm base plug & stacking socket

DUAL-STAGE FACTORING LID & L-RAILS

Slide L-rails into lid to set Cartesian origin.
Align tiles against rails to read binomial factors!

AREA MODEL: (x + 3)(x + 2) = x² + 5x + 6



SOLID RECTANGLE

Zero gaps, zero overlaps
Linear: 3x + 2x = 5x
Constant: 3 × 2 = 6

Un-multiplying area!

2. KIT PARTS INVENTORY (34 PIECES)

COMPONENT	QTY	SPEC / DIMENSION	PALETTE COLOR
Quadratic (+x²)	1x	35.00 × 35.00 mm (5U×5U)	Emerald Green
Quadratic (−x²)	1x	35.00 × 35.00 mm (5U×5U)	Royal Purple
Linear Rod (+x)	5x	35.00 × 7.00 mm (5U×1U)	Emerald Green
Linear Rod (−x)	5x	35.00 × 7.00 mm (5U×1U)	Royal Purple
Constant (+1)	9x	7.00 × 7.00 mm (1U×1U)	Warm Gold
Constant (−1)	9x	7.00 × 7.00 mm (1U×1U)	Crimson Red
Corner L-Rails	2x	105.0 mm Cartesian Axis	Slate Gray
3x1 Storage Bin	1x	49.0 mm Height (7U)	Matte Charcoal
Stage Lid	1x	Dual-Stage Factoring	Matte Charcoal

TOTAL: 30 Math Tiles + 2 L-Rails + Storage Bin + Stage Lid = 34 Parts

DUAL-COLOR SIGN CONVENTION (+ / −)

- Emerald / Gold: Positive mathematical values (+x², +x, +1).
- Royal Purple / Crimson: Negative values (−x², −x, −1).
- Zero-Pair Elimination: Positive cancels negative on sight.
- Matte Texture: Anti-glare finish for document cameras.

3. WHY KEEP THIS REFERENCE CARD?

✓ NO LOST PIECE LEFT BEHIND

Classroom manipulatives get lost. Never abandon an incomplete set! Every individual math component is permanently stocked for single replacements.

Scan back QR or visit stackcalc.io/parts
Order individual replacement pieces anytime

✓ 10-SECOND SELF-CHECKING PACK-UP

Pack-up is a built-in math game! Lay the two x² tiles flat on the box floor, slot the 10 x-rods in side bays, and stack the 18 unit cubes in center bays. L-rails snap securely into lid retention. Lid latches firmly for silent, rattle-free carry.

✓ HANDS-ON TUTORIAL ON REVERSE

Flip this card over for a complete step-by-step hands-on lesson with concrete numbers, stage arrangements, and calculator cross-references!

Keep this card stored inside the box lid!

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California, USA • Non-Toxic BPA-Free.
Classroom Lab Crates (10x / 30x Stations): stackcalc.io/schools

FUNDAMENTAL THEOREM OF ALGEBRA

Every degree-n polynomial has exactly n complex roots counted with multiplicity.

Universal 7.00mm Datum • Classroom Continuum (10 Kits)

OFFICIAL REFILL PORTAL

stackcalc.io/parts
Order replacement parts anytime

SERIES 1 • CARD 06 OF 10 • DOMAIN: POLYNOMIAL GEOMETRY & BINOMIAL FACTORING • CARTESIAN L-RAILS • ZERO-PAIR TEXTURE CONTRAST

LESSON GUIDE: FACTORING TRINOMIALS & BINOMIAL ARRAYS

Step-by-Step Hands-On Tutorial: Factoring $x^2 + 5x + 6 = (x+2)(x+3)$

Grades 7–11 Pre-Algebra & Algebra II • Hands-On Classroom & Home STEM Studio Series

CARD 06 / 10 • ALGB-01

MATHEMATICAL GUIDE

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/algebra-boxes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS.MATH.CONTENT.HSA.SSE.A.2 (Structure)
- CCSS.MATH.CONTENT.HSA.APR.A.1 (Arithmetic)
- TEKS 111.39.c.10.E (Factoring Trinomials)

Grades 7–11 Pre-Algebra & Algebra II

MATHEMATICAL INQUIRY

Can you arrange the linear rods so the rectangle closes completely without any gaps or overlapping edges?

1 STEP 1: SEAT ORIGIN TILE (+x²)

Slide the two L-rails into the lid corner to form the origin.
Place the emerald +x² tile flush into the corner.
Its 35.00 × 35.00 mm area establishes the quadratic core.

2 STEP 2: DISTRIBUTE 5x LINEAR RODS

We need to distribute 5 positive x rods around the +x² tile.
Place 3 rods along the top rail (horizontal dimension).
Place 2 rods along the left rail (vertical dimension).
Notice: 3 + 2 = 5 linear terms accounted for!

3–4 STEPS 3 & 4: COMPLETE RECTANGLE & SOLVE

3. COMPLETE THE RECTANGLE: Fill missing corner with 6 constant (+1) tiles!
4. READ BINOMIAL DIMENSIONS: Horizontal width is (x + 3); vertical length is (x + 2).
Total Area: $(x + 2)(x + 3) = x^2 + 5x + 6$!
Calculator Verification: Press Polynomial Solver Icon ([↱][EQN] & [↱][POLY]) and run [↱] [POLY] 1 [A] 5 [B] 6 [C] 0 [D] [EXEC] [RCL] 1 [RCL] 3 → −2.

5 CORE PRINCIPLE: FACTORING IS UN-MULTIPLYING AREA

Factoring a quadratic is literally finding the length and width of a rectangle! If the tiles cannot form a solid rectangle without gaps, the trinomial is prime. Zero-pairs (+x and −x) let you model differences of squares $x^2 - 4$ with effortless physical cancellation!

★ EXTENSION INVESTIGATION: FACTOR: x² + 6x + 8

Arrange one +x² tile, six +x rods, and eight +1 units into a solid rectangle. What are its rail dimensions? (Scan QR for 3D animated solution!)

→ Assemble with your physical blocks, then scan the QR code to verify with the 3D step-by-step vector model.

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Interactive Mathematics Manipulative Ecosystem.
Designed in Mountain View, California • Archival BPA-Free POM/PETG.

CROSS-REFERENCED CALCULATOR VERIFICATION

Every keystroke sequence verified with RPNCore CalculatorEngine.
Modular Hardware Continuum (9 Kits) • Universal 7.00mm Pitch • stackcalc.io

PARTS & SCHOOL REFILL PORTAL

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